Assignment: Noise Schedule Ablation in Diffusion Models
Phase 1 — Declaration

Project: Underwater image enhancement with the pretrained **UW-DiffPhys** model, fine-tuned under the original and a power-transformed noise schedule

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1. Image-to-image task: Requirement is to state the image-to-image task which is chosen by the group

Paired underwater image enhancement. Given a degraded underwater RGB image x_0 (colour cast, haze, low contrast), generate the corresponding enhanced reference image y_0. The mapping $x_0 \rightarrow y_0$ is learned by a conditional diffusion model on paired data, with 128×128 inputs and outputs (the native resolution of the pretrained model; EUVP's 256×256 images are resized).

2. Pretrained model: Requirement is to link to the paper or codebase of the chosen pretrained model.

UW-DiffPhys — Underwater Image Enhancement with Physical-based Denoising Diffusion Implicit Models

Paper	https://arxiv.org/abs/2409.18476
Codebase	https://github.com/bachzz/UW-DiffPhys/tree/main
Checkpoint	LSUI_UIEB_ddpm.pth.tar, linked from the codebase README (Checkpoints section): https://drive.google.com/file/d/13dZWgUW7tCgBf4SVvqoCZUAHniKcMiU8/view?usp=sharing

3. Dataset Requirement: Dataset name and Source URL.

EUVP — Enhancing Underwater Visual Perception (paired subsets), Islam, Xia and Sattar.

Source (Kaggle)	https://www.kaggle.com/datasets/pamuduranasinghe/euvp-dataset?resource=download
Original release	https://irvlab.cs.umn.edu/resources/euvp-dataset
Dataset paper	https://arxiv.org/abs/1903.09766
Subsets used	Underwater Dark (5,550 training pairs, 570 validation) and Underwater Scenes (2,185 training pairs, 130 validation), counts as listed on the original release page. The Underwater ImageNet subset is excluded.

4. Evidence that the dataset was not used in pretraining

Requirement:

Option (a). A link to the paper or codebase, with page number and line number, confirming the dataset is absent from the pretraining data.

Option (b): A link to the paper or codebase, with page number and line number, describing the pretraining data domain, and a one-paragraph argument that the chosen dataset belongs to a clearly different domain

Source 1 — paper	https://arxiv.org/abs/2409.18476

Section IV-A, “Experimental Methods”, item 2 “Datasets and Evaluation Metrics”: page __, lines __–__.

The model is trained on LSUI-UIEB-Train, i.e. UIEB-Train (800 pairs) combined with all 5,004 LSUI pairs. The other datasets named there (UIEB-Test90, UIEB-Test60, U45-Test) are used for testing only. EUVP is not listed anywhere in the paper's training or evaluation data.

Source 2 — codebase	https://github.com/bachzz/UW-DiffPhys/tree/main

README.md: the “Data” section lists LSUI-UIEB as the data, and the “Training” command uses config underwater_lsui_uieb_128.yml with checkpoint LSUI_UIEB_ddpm.pth.tar: page n/a, lines __–__.

EUVP sources	EUVP images come from seven cameras used by its authors, public YouTube videos and, for the excluded ImageNet subset, ImageNet and Flickr images (EUVP paper, Section 3.3: https://arxiv.org/pdf/1903.09766). UIEB, LSUI and U45 are not listed as sources. Excluding the web-sourced ImageNet subset further reduces the chance of overlap with the web-collected UIEB and LSUI images.

5. Native noise parameterization: Requirement is to identify which quantity serves as θ, and confirmation that θ ∈ (0, 1) for all t.

θ = $\bar{\alpha}_t$, the cumulative product of the per-step retention coefficients (paper Eqs. 7–9, Section II-C-1).

Forward process: $q(x_t \mid x_{t-1}) = N(\sqrt{\alpha_t}\, x_{t-1}, (1-\alpha_t)I)$, with $\alpha_t = 1 - \beta_t$; closed form $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{(1-\bar{\alpha}_t)}\, z_t$, $z_t \sim N(0, I)$, where $\bar{\alpha}_t = \prod_{i=1..t} \alpha_i$, t = 1, …, T. Training (Eqs. 22–27) and DDIM sampling (Eqs. 18, 30) use $\bar{\alpha}_t$ directly.

Confirmation θ ∈ (0, 1) for all t. The paper states $\alpha_t \in (0, 1)$ (Eq. 7 text). A finite product of numbers in (0, 1) lies in (0, 1), so $\bar{\alpha}_t \in (0, 1)$ for every t. It is also strictly decreasing, since $\bar{\alpha}_t = \bar{\alpha}_{t-1}\, \alpha_t < \bar{\alpha}_{t-1}$. The exact β_t range and T are read from underwater_lsui_uieb_128.yml and printed in the run logs; sampling uses a 25-step DDIM sub-sequence of the T training steps.

Modified schedule. $\{\theta^x\} = \{\bar{\alpha}_t^x\}$ also lies in (0, 1) and decreases strictly with t; its per-step form is $\alpha'_t = \bar{\alpha}_t^x / \bar{\alpha}_{t-1}^x = \alpha_t^x \in (0, 1)$, so no conversion to another parameterization is needed.

6. Chosen exponent x: Requirement is to choose exponent x to two decimal places, x ∈ (1.5, 3.0).

x = 2.00 (within the allowed interval (1.5, 3.0)). Run B uses the schedule $\{\bar{\alpha}_t^{2.00}\}$, applied elementwise.